LEGAL PROBLEMS POSED BY THE COMMERCIALIZATION OF DATA COLLECTED BY THE EUROPEAN REMOTE SENSING SATELLITE ERS-1

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Satellite remote sensing is one of the oldest and most widely used space applications. Like meteorology, it pertains to Earth observation activities which include the study of the globe (soil, subsoil, oceans) and its resources (ores, hydrocarbons, agriculture, forests, fishing). It calls forth several scientific disciplines (geodesy, geology, hydrology, climatology). It has very extensive spin-offs in every field (political, economic and military).

Like other space activities, satellite remote sensing ignores the natural and artificial frontiers of the different States. This entails consequences particularly important, because the States can be observed unknowingly and against their will. Furthermore, the so-collected information can be used or disseminated without their knowledge and thus benefit the observing State as well as third States without the observed States being able to benefit by this information. This situation is aggravated by the fact that the data collected by satellites can be used only after undergoing a complex processing on the ground which requires specialized equipment and technical training not easily accessible to all countries.

In legal terms, this means that satellite remote sensing is, like radio broadcast from space, an activity which poses the difficult problem of reconciling the principle of free exploration and use of outer space with the principle of the States' sovereignty over their territory and natural resources.

The problem is still more complicated by the tendency, which develops continuously in certain States, to give up performing satellite remote sensing activities by themselves and leave them, either as a whole or only in part, to the initiative of nongovernmental organizations or entities. This process is sometimes referred to as "commercialization".

Commercialization is already in existence in several countries:

- in the United States, the "Land Remote Sensing Commercialization Act" dated 17 July 1984 provided for the constitution of a private remote sensing system and the Eosat company was subsequently granted a license for operating Landsat satellites;

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- in France, the Spot-Image company was established in late 1981 with a view to using the data collected by the Spot satellite and selling them to the public, an activity which it has pursued since 6 May 1986. A Spot-Image subsidiary was established in the United States under the name of Spot-Image Corp (SI Corp.);

- in the USSR, the Soyouzcarta Company broadcasts the products of Soviet satellites, to an area including select Western countries (Europe, Australia, United States).

Harsh competition opposes these various companies, particularly as regards the photograph definition. This varies from 30 m with the Americans, 10 m with the French, to 5 m with the Soviets. In order to take these differences into account, the United States has just decided to put satellite photographs of "military quality" at the disposal of the public - except where otherwise justified for national security reasons.

Lastly, we must also mention the existence of a project for an Earth observation satellite to be developed jointly by China and Brazil which must be viewed as a commercial rival of the above-mentioned satellites.

In an attempt to find a solution to this difficult and multi-faceted problem, the United Nations Organization has, for more than ten years, made considerable efforts which resulted, not without great difficulties, in the adoption in December 1986 of a Resolution on the principles which should henceforth govern satellite remote sensing activities.

As pointed out before, these activities were first performed by the different States in fulfilling objectives of their own, which were part of their mission to protect the interests of all their nationals.

At present time, remote sensing satellites are built, launched and used in orbit by the States themselves, but in some of these States the data collected by these satellites are processed and disseminated by nongovernmental organizations or entities. The development of "commercialization" will result in all such activities, whether in space or on ground, being performed by such organizations or entities.

It is within this framework that the Programme called "ERS-1" (European Remote Sensing Satellite N 1) must be placed. This Programme is the first in the field to be undertaken by several European countries (joined by Canada). The consequences of the association of several States for this undertaking increases the difficulty of the political and legal problems we have mentioned.

This is why it seems appropriate to study, from a legal viewpoint, the conditions for the consequences of the "commercialization" of data collected by the European Remote Sensing Satellite ERS-1.

To this end, it is advisable to discuss initially the general legal aspects of the problem, before describing those specific to the ERS-1 programme.

I. GENERAL LEGAL ASPECTS

The general legal aspects of the problem posed by the commercialization of data collected by the European satellite ERS-1 are, first of all, those applicable to every remote-sensing progamme, *i.e.* the compliance with the general principles which govern remote-sensing activities. It is also advisable to consider the principles applicable to the commercialization of space activities.

A. General principles of remote-sensing activities

The first principles governing the execution of remote sensing activities were laid down in international law by the agreement signed in Moscow on 19 May 1978 on the transfer and use of data resulting from Earth remote-observation from Outer Space.[1]

This agreement binds the members of Intercosmos, an agency for scientific co-operation among Eastern countries, the aim of which is, in particular "to study the natural environment (of the Earth) through the use of space objects." However, it binds only those who are parties and tends to settle only the situations created within the framework of co-operation organized among Intercosmos members.

The general problem concerning the activities of Earth remote observation from Outer Space could be solved only within the largest existing framework, namely that of the United Nations. As mentioned earlier, it took more than ten years of discussions within the Space Committee before the General Assembly of the United Nations could adopt on 4 December 1986, Resolution 441/65 which sets forth these principles.

In fact, the Resolution uses the term "remote-observation" only in its title while, in the text itself, it uses the term "remote-sensing" which is defined in Principle I(a) as "[t]he observation of the Earth surface from space, using the properties of electromagnetic waves emitted, reflected or diffracted by the observed bodies, for the purpose of improving the management of natural resources, parcelling out the territory and protecting the environment." Sub-paragraph (e) of the same Principle defines remote-sensing activities as "[t]he activities consisting in operating remote-sensing satellites, primary data receiving and filing stations as well as the data processing and processed data interpreting activities." These technical expressions "primary data", "processed data", as well as the term "analyzed information" are also defined in subparagraphs b, c and d of Principle I of the Resolution. This would

1. The following countries are parties to the Moscow Convention of 19 May 1978: the German Democratic Republic, Cuba, Mongolia, Poland, Rumania, Czechoslovakia and the USSR. Gál, *Legal Principles of Remote Sensing in the Moscow Convention of May 1978,* in PROC. 38TH COLL. L. OUTER SPACE 2 (1987).

indicate that the authors of the text were very anxious to set the Principles on a strong and realistic technical basis, not on theoretical considerations of a political or legal nature. On the other hand, it should be noted that the scope of the Resolution does not cover only the "outer space" part of remote-sensing activities, but also their "ground" part which, incidentally, seems to strengthen the point of view of those who are in favour of a "functional" instead of "physical" definition of outer space.

This article is not intended to comment on the fourteen other Principles included in the Resolution, the aim of which is, if not to resolve the conflict between the notion of freedom and that of sovereignty, at least to propose a compromise as regards their application.[2]

However, it must be kept in mind that a United Nations' Resolution cannot be considered as affording a final solution to this conflict since the legal value and the mandatory character of such texts are, at least, controverted.

Moreover, we know that recourse to the entry of "Principles" in a Resolution occurs only as a last resort where members of the UN Outer Space Committee have not been able to agree on the idea of preparing a Treaty.

However that may be, it is obvious that the Principles entered in the Resolution - which was adopted unanimously by the General Assembly - should be applied by every State, since they have become an integral part of the international law which Article III of the Treaty dated 27 January 1967 compels these States to observe. There must be no doubt about that as regards the States participating in the ESA, ERS-1 programme, which are all Parties to the Outer Space Treaty.

B. General principles on commercialization

One of the fundamental rules laid down by the Outer Space Treaty is the affirmation of the States' international responsibility for the fulfilment of space activities, whether these activities are undertaken "by governmental agencies or by non-governmental entities". Under Article VI of the Treaty, such activities must be authorized and continuously supervised by the appropriate State party to the Treaty. This article thus acknowledges the lawfulness of the space activities undertaken by non-governmental entities or organizations - activities usually referred to as

2. The United Nations' Resolution 41/65 of 4 December 1986 contains 15 Principles. Principle I gives a definition of Remote Sensing activities. Principles II, III and IV lay down the general principles of international law which must be used as a basis by the States which carry out Remote Sensing activities. Principles V to IX contain an incitement to international co-operation. Principles X and XI aim at protecting the environment and preventing natural disasters. Principles XII and XIII relate to the rights of the observed States. Principle XIV affirms the international responsibility of the States and Principle XV deals with the conciliatory settlement of disputes.

"commercial activities." But these activities are subject to the authorization and continuous supervision of the State to which the entity or organization in question is answerable. The State must ensure that "national activities are carried out in conformity with the provisions" of the Treaty, thus stating its international responsibility.

Last, it must be kept in mind that Article VI stipulates that in case of activities carried out by an international organization, the responsibility for compliance with the provisions of the Treaty will rest with this international organization and the States parties to the Treaty which belong to the aforesaid organization.

Thus, for the above-mentioned reasons, all remote-sensing activities which can be referred to as "commercial activities" must be fulfilled in compliance with the principles laid down in the United Nations Resolution dated 4 December 1986.

It should be noted that this is true whatever the nature of these activities, *i.e.* whether they take place in space or on earth. This is true whether these activities are carried out directly by a State or an inter-governmental organization, or by a nongovernmental organization or entity under responsibility of the appropriate State.

II. LEGAL BASES OF THE ERS-1 PROGRAMME

The ERS-1 is a European Space Agency programme which allows the satellite it intends to build and operate to be called a "European" satellite. But this term must not be given a strictly geographical meaning. It is used only because the programme is carried out under the conditions specified in the Convention providing for the establishment of a European Space Agency.[3]

One of the characteristics of this Convention is that it makes it possible for the Member States - which must mandatorily participate in a scientific programme - to carry out, jointly and within the framework of the Agency, optional programmes in the field of space applications. These optional programmes are carried out according to the rules laid down by the Convention itself in their main lines, but likely to be adapted to particular circumstances.[4]

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3. The Convention establishing a European Space Agency was signed on 30 May 1975. It came into force on 30 October 1980.

At the present time, the European Space Agency (ESA) includes 13 Member States: the Federal Republic of Germany, Austria, Belgium, Denmark, Spain, France, Ireland, Italy, Norway, the Netherlands, the United-Kingdom, Sweden and Switzerland, and well as an Associated State, Finland. Canada is bound to ESA by a co-operation agreement and participates in several optional programmes.

4. Article V.1 (b) and Annex III of the Convention establishing the European Space Agency. For more detailed information about the optional ESA programmes, see Bourély, *Institutional Arrangements for Space Cooperation in Europe*, in PROC. 24TH COLL. L. OUTER SPACE 159 (1981).

Such is the case with the development of the ERS-1 satellite. It is advisable to outline the technical aspect of this optional programme before discussing the legal rules with which it must comply. We will intially consider only the "R & D" aspect of the programme, since the use of data collected by the satellite falls under different rules which will be discussed in the third and fourth parts of this Article.

A. General framework for optional programmes of ESA

The system dealt with by the Convention is based on the fundamental idea that optional programmes are programmes specific to the Agency itself. Thus, the Agency supplies the framework in which they are implemented, using the necessary personnel and facilities (establishments and general-purpose installations). On the other hand, these programmes are financed only by those member States desirous of doing so. These are called the "Participants". Hence it follows that only the latter will have a say in the decision-making process concerning these programmes.

To implement this dual principle, the Convention provides for the involvement of three legal instruments:

(a) A *Resolution of the Council* (called "Empowering Resolution" by which the Board agrees that the planned optional programme will be implemented "within the framework of the Agency");

(b) A *Declaration* subscribed by the ESA member States desirous of undertaking the framework of an optional programme jointly and within the Agency. The draftsmen of the Convention proposed to encourage such programmes by stipulating that each member State is to participate in each optional programme, unless it has positively specified that it is not interested. Each Declaration contains a number of articles describing the Participants' commitments. It is complemented by two appendices, a technical one giving a more or less detailed description of the programme, its objectives, its timescale, its phases, *etc.*, and a financial one fixing a budget allocation for the implementation of the whole programme and determining the scale of contributions. In this respect, two important points must be made:

- the budget allocation is an estimation, but it is binding on the Participants. However, the Participants agree that it can be increased by 20% without the programme being questioned. If the possible overdraft is to be greater than 20%, each Participant will have a right to withdraw from the programme;

- on principle, contributions are calculated on the basis of the Participants' national income, but the Participants are entitled to agree to any other scale for the saving of expenses.

(c) *Implementing Rules* adopted by the Participants, containing the detailed methods agreed to by the Participants for implementing the programme. In particular, the necessary decision-making procedures

(appointment of the competent deliberative body, voting rules), as well as the application - subject to express exceptions - of ESA rules to contractual, financial and other matters. These Rules stipulate the conditions under which some non-member States may be allowed to join the Participants with a view to implementing the programme in question.

It should be pointed out that the Empowering Resolution is voted by the Council (by a simple majority of votes) *i.e.*, by all member States, whereas the Declaration - which is prepared and subscribed unanimously by all Participants - is submitted to the Council only for information. As regards the Implementing Rules, also prepared by the Participants, they must be submitted for approval to the Council whose decision is made by a simple majority of votes.

B. Technical aspect of the ERS-1 programme

Until quite recently, the European Space Agency had not had a remote-sensing satellite of its own and had merely been acquainted with this field by collecting and disseminating in Europe the data supplied by US satellites, particularly Landsats, through its "Earthnet" network.

In March 1982, ESA was authorized to initiate its first programme intended to develop and build remote-sensing satellites called "Earth Resources Satellites" (ERS). These will constitute the third generation of satellites of this kind in the Western World. They will complement Landsat (US) and Spot (France) satellites and will contribute to a better monitoring of the environment of the Earth and the exploitation of its resources. They will have an advantage over their predecessors in that they will be equipped with a system of ensuring a quick transmission of data whose processing and delivery will be guaranteed within three hours.

The objectives selected for ERS-1, the first satellite of a series, tend mainly to facilitate the exploitation of coastal oceanic areas (particularly the ice-packed areas), on the one hand, and the development of an improved meteorological information system at the earth scale, through the collection of data on weather conditions above the oceans, on the other. Lastly, ground objectives will be studied by means of an active VHF instrument (AMI) with two operating modes: either in the wind diffusion-metering mode or in the synthetic aperture radar (SAR) mode permitting to scan obliquely a 100-km wide strip of the satellite trace on the ground.

Two other instruments, provided by several participant States, will be added to those provided by the space agency: first, the most accurate scanning radiometer along the path (ATSR), presently available, providing sea surface temperatures, and measuring equipment for distance and speed (PRAREA). The States providing these instruments are financing the associated costs.

Owing to its remote-sensing systems, ERS-1 will enable scientists to follow the movement of oil slicks and icebergs, to study the effect of

acid rains on forests and lakes, the influence of ground erosion on farmlands and, also, to detect areas of geological faults in order to locate places at risk of earthquake. With the use of the data transmitted to the ground, it will be possible to locate continental - and to some extent submarine - ore deposits with accuracy. It will also be possible to locate spawning areas and to follow the growth of cultures.

The economic interest of these various applications is thus obvious and it will be increased to a great extent by the presence of quick data transmission systems on board the satellite.

ERS-1 is currently under construction by the European industry. Its launch by Ariane is scheduled to take place in early 1990 and it will have a two-year life expectancy. The construction of a second satellite, ERS-2, is envisioned and a launch in 1993 is planned.

Lastly, it is worth mentioning that the total cost upon completion of the ERS-1 programme (including launch costs) was estimated to be about 373 million European Currency Units (1 ECU =1 US dollar) in 1982.

C. Legal instruments constituting the ERS-1 programme

All ESA optional programmes comply with general rules laid down by the Convention and complemented by specific provisions. Thus, the ERS-1 programme fits quite naturally with this general framework, within which some adaptations have been made.

The adoption of the various legal instruments relating to the ERS-1 programme took place in successive steps.

At a meeting of the Agency Council, held at ministerial level in February 1977, a Resolution was adopted, placing emphasis on the interest of Europe to engage in a preparatory remote-sensing programme.

By a Resolution of 28 October 1981, the Council agreed to the implementation of a European remote-sensing satellite (ERS-1) programme with the Agency.

The Declaration relating to this programme was made out by the Participant States on 24 March 1982, updated on 16 June 1982 and amended on 19 July 1983. It was complemented by Implementing Rules established July 28, 1983, and approved on 27 May 1982 and amended in October 1983.

The following States have adopted the Declaration and are therefore Participants in the ERS.1 programme: the Federal Republic of Germany, Belgium, Denmark, Spain, France, Italy, Norway, the Netherlands, the United Kingdom, Sweden and Switzerland. Furthermore, Canada, although not an ESA member State, is a Participant in the programme.

The successive steps gone through by these legal instruments resulted from the fact that the Participants wished initially to commit themselves only for a definition phase (Phase B). An additional Declaration dated 11 July 1984, made by the Participants themselves

within the framework of a "Programme Board" allowed the next phases to be initiated (Phases C-D: system development and validation; phase E: preparation and execution of system operation).

Two annexes (A and B) are attached to this Declaration. They update the Technical and Financial Annexes of the initial Declaration. The additional Declaration contains also an Annex C which incorporates the principles agreed by the Participating States in respect to access to the ERS-1 System and of distribution of the resulting data and products. This additional Declaration was updated on 18 September 1985.

At a meeting held in Rome on 31 January 1985, the Ministers decided to "carry on ESA activities sturdily in the field of Earth observation," focusing them on the already-approved ERS-1 project and on future elements, namely, further missions of ERS-1 oceanographic and meteorological applications, a ground application project, the participation in the development of a second-generation meteorological satellite, and mission studies of solid globe physics, the atmosphere and climatology.

These directives were confirmed by the Ministers at a new meeting at The Hague on 9-10 November 1987. They affirmed the objective consisting of: "anticipating a substantial contribution of space and ground techniques to the Earth observation sciences and their applications and preparing, to the extent that it may be required, the setting-up of operational systems and organizations centered on the users for the operation of these systems. The Ministers have therefore stressed the interest for Europe of realizing a fair balance between infrastructure programmes (Ariane 5, Hermes, Columbus, Data Relay Satellite) and utilization programmes, in particular the observation of Earth resources. Moreover, the Ministers considered that the efforts currently made by Europe were: "a source of new possibilities for the private sector which should be encouraged to use the available potential, to participate in investments and assume responsibilities for the operation of such systems."

III. LEGAL ASPECTS OF THE OPERATIONAL PHASE OF THE ERS-1 PROGRAMME

Without elaborating any further on the characteristics of the ERS-1 programme as an optional development programme of the European Space Agency, it is now necessary to consider the legal problems posed by the operational phase of this satellite and to set forth the specific rules agreed to by the States participating in the programme. Additionally, the answer the States have given to the question of intellectual property is touched upon.

A. Specific rules laid down within the framework of the ERS-1 programme

As stated in the initial Declaration, the ERS-1 programme includes a so-called "operational" final phase (phase E) the nominal duration of which is two years. As a result, unlike other ESA programmes in which the developed equipment becomes the responsibility of a user organization once the programme is completed, the Agency itself will at least at the beginning, assume responsibility for the operation of the satellite and the dissemination of its data.[5]

Conscious of the problems that might be raised by this situation, the States participating in the programme have laid down special provisions for the operational phase (phase E). As in the other phases of the ERS-1 programme, the Declaration details the respective tasks and responsibilities of the Agency, and the participating States during this phase.

1. Role of the Agency during the operational phase

As regards the operational phase of the programme, Article 2 of the Implementing Rules stipulates that the Agency "fulfills or co-ordinates" the following tasks:

- raw data recording;
- data pre-processing and processing in quasi-real time, mainly for mission control and management purposes;
- data archiving, preparation and keeping of catalogues;
- elaboration of thematic products in real time;

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5. The transmission of equipment developed by the European Space Agency to organizations in charge of its commercial use takes place at the present time in the following file:

- Launch vehicles for the benefit of Arianespace, a "commercial" company under the French law;
- Telecommunication satellites, for the benefit of Eutelsat, and inter-governmental organization;
- Meteorological satellites, for the benefit of Eumetsat, and inter-governmental organizations.

Moreover, in the field of Remote Sensing, the dissemination of data from American satellites (Landsat), connected as part of the Earthnet programme, has been entrusted since 1 January 1987 to a European group of manufacturers, in Eurimage. The establishment of these various organizations intended to assume responsibility for operational activities was agreed to jointly with ESA. However, the ESA Convention (Art. V.2(c)) enables ESA to perform "operational activities" itself such as the operation of the equipment developed by ESA, if the users ask it to do so, which is the case with the ERS.1 operational phase.

- data and product transmission and dissemination;
- mission control and management;
- demonstration of applications.

Moreover, "the Agency remains in contact with national Centres having a thorough knowledge in the field of remote sensing, particularly as regards the elaboration of thematic products in quasi-real time and the demonstration of applications."

Article 4 stipulates that the Agency:

- provides for the installation of the ground sector of the programme;
- determines the methods of data dissemination in quasi-real time to the main user centres of the participating States;
- coordinates the use of the Satellite by all acquisition stations which have access to the ERS satellite as well as the processing of acquired data;
- make necessary arrangements with the participating States as regards the use of their data processing facilities, on common bases ensuring similar conditions for all the participating States.

Lastly, Article 6 of the Regulation entrusts the Agency with the task of placing the contracts necessary for the implementation of the programme.

The institutional organization set up for the operational phase thus duplicates that of the development phase to the extent that the Agency is charged by the participating States to fulfill, on their behalf, a number of functions which would normally be incumbent on them.

As regards the operational phase of the ERS-1 programme, the Agency has designed the architecture of the data processing and the data archiving in a manner that will satisfy the exigencies of a vast community of users, from operators in real time to research groups.

The configuration kept for the ground sector revolves around a central installation set upon the ESA establishment in Italy (ESRIN) where the starting point of the Earthnet network is found. It includes, Processing and Archiving Facilities, or PAF, established in the Federal Republic of Germany, France, Italy and the U.K.

The principal functions of the PAF are:

- long-term arching and processing of raw data and some ERS-1 subsidiary information;
- the creation of geophysical products and autonomous precision, as well as their dissemination;
- the support for the evaluation of long term performances of remote sensors, the calibration and geophysical validation of remote sensors, the demonstration campaign, and pilot projects;
- the interface with the ERS-1 central installation of ERSIN which will be charged with the coordination of the ensemble of PAF installations.

2. The Role of Participating States in the Operational Phase

As a counterpart of the missions they ask the Agency to effect on their behalf, the Participating States take on the commitment to contribute to the financing of the different phases of the program and, in particular, of the operational phase, according to specific rules (Art. 5.2). They also agree to support the financial cost of all obligations that the Agency should incur as a consequence of its international responsibility (Art. 9.1).

However, the Participants reserve the right to exercise a number of prerogatives which are valid for the whole programme:

- the general rules of the Agency in terms of geographic return must be complied with (Art. 6);
- the rights of intellectual property, including access, communication and utilization rights deriving from inventions and technical data resulting from contracts and sub-contracts placed by the Agency for the implementation of the programme, are reserved for the benefit of the participating States and for the benefit of the Agency (Art. 7.1);
- the goods produced within the framework of the programme, as well as the facilities and equipment purchased for its implementation, are the property of the Participants, through the medium of the Agency (Art. 8).

B. Rules applicable to the intellectual property regime

1. General aspect of the problem

As we know, the regime of intellectual property applicable to the data collected by Remote-Sensing Satellites poses very specific problems. First of all, can we really say that the data collected by such satellites can be protected as being subject to the various intellectual property regimes existing throughout the world? If so, which system must be referred to? The regime of the State which launched the satellite, or that of the State where raw data is collected by a receiving station? But as raw data can be used only insofar as it has been processed more or less extensively, what sort of protection must be granted to it?

These questions, which are much discussed in the doctrine, find no answer in the United Nations' Resolution 41/654, the aim of which is only to lay down principles on how to carry out Remote-Sensing activities, and not to solve these kinds of difficulties.

International agreements on the protection of copyrights and the few national laws in existence do not enable us to meet fully and satisfactorily the necessity of properly protecting the legitimate interests of those who receive and process remote-sensing data with a view to

commercializing it. Thus, sooner or later, it will be necessary to work out an international agreement on that subject.[6]

2. Principles adopted within the framework of ERS-1 programme

The states participating in the ERS-1 programme did not pretend to solve the problem as a whole but they have adopted, in the Implementing Rules and for the needs of this programme, some provisions which deserve to be mentioned.

(a) First of all, Article 7.1 deals with intellectual property concerning the programme proper. This text stipulates that "when placing contracts and sub-contracts for the implementation of the programme, the Agency reserves, for its own needs and those of the participating States, as regards the resulting inventions and technical data, the intellectual property rights, including the rights of access to and communication and use of technical data."
- Second, "the intellectual property rights resulting from experiments financed by virtue of the programme are reserved for the participating States. However, the Agency, acting on behalf of the participating States, is the owner of these rights." (Art. 7.2).
- Third, "the results of experiments conducted in the course of the programme are communicated according to the rules laid down by the Agency concerning the scientific and technical information." This clause, which exists also in the ESA programmes of a scientific nature, is yet subject to a restriction as regards ERS-1, namely: the participating States or the national organizations answerable to them reserve the exclusive right to publish the results concerning the operation of the instruments supplied by them and of their experiments for a period varying from 6 to 18 months. (Art. 7.3). Moreover, no result concerning the in-flight operation of an instrument can be published without the express agreement of the State or organization that has financed it. (Art. 7.3b and c).
- Fourth, "the intellectual property rights resulting from the use of additional instruments belong to the organizations that have financed these instruments. Nevertheless, the Agency is entitled to use them free of charge for all its activities and programmes." (Art. 7.4).
(b) The above clauses relate to the results of the ERS-1 programme development phases. As regards the data collected during the satellite operations phase - a question which is directly linked to that of commercialization - Article 7.4 reads: "Also reserved to

6. The protection of Remote Sensing data was alluded to in several recent IISL colloquia. *See*, in particular, R. Oosterlink, *Legal Protection of Remote Sensing Data* in PROC. 27TH COLL. L. OUTER SPACE 112 (1985).

the participating States are the access to technical information, raw data and thematic products obtained during the operational phase of the programme, and the right to use it for the needs specific to the participating States." Thus, this wording, which sets the notion of "access" against that of "use," does not mean that the participating States do not wish to take it upon themselves to disseminate the data in its various forms. In fact, and as it is pointed out in the continuation of Article 7.4, "the Agency is in charge of co-ordinating the dissemination and management of the ERS-1 elaborated products in accordance with the data dissemination policy laid down by the programme Steering Board."

When we read the above text, two observations come to mind: the first one is that the Agency is thus entrusted with an additional mission in a field where it must substitute itself - at least temporarily - for the users' community. The second is that the programme Steering Board must lay down the data dissemination policy, a task which it has not yet fulfilled and which should be the subject of an addendum to the programme Implementing Rules.

IV. ERS-1 DATA DISSEMINATION

As pointed out earlier, Appendix C to the additional Declaration of 11 July 1984 lays down the principles governing access to the ERS-1 system as well as data and product dissemination. These principles are implemented as stipulated in the programme Implementing Rules.

As a number of questions still remain to be settled among the Participants, particularly the price policy, we will limit ourselves to a brief description of the structure that will be used for the dissemination of ERS-1 data and products. This structure will include three levels:

1. The first level is that of ground reception stations. These are, first of all, the stations belonging to the national organizations of the participating States. Article 4 of the programme Implementing Rules provides that "the Agency determines the methods to be used for data dissemination in quasi-real time to the main user Centres of the participating States." It "co-ordinates the use of the satellite by every acquisition station having access to the ERS satellite as well as the processing of the acquired data." Lastly, "the Agency makes arrangements with the participating States for the use of their data processing facilities on common bases ensuring similar conditions for all participating States." (Art. 4.4).

There may also be some reception stations which do not belong to the participating States, particularly those designed for receiving ERS-1/SAR data. They will be invited to sign an agreement with ESA on "direct reception, archiving and dissemination of ERS-1/SAR data." This

agreement will bear on the definition of the clauses and conditions applicable to:

- the direct access to ERS-1/SAR data by the ground station;
- the putting of the ERS-1 data acquired by the ground station at the disposal of the users' community;
- the supply of LBR data to ESA for acquisition in real time in case of failure of ERS-1 on-board tape recorders.

The agreement will also contain an article acknowledging ESA's right to the property of all ERS-1/SAR data, on the one hand and ESA's copyright on the ERS-1/SAR data, on the other hand and consequently the ESA's rights to this data. A license will be granted to the station by the Agency but, as a reciprocal measure, ESA will have the right to look into the dissemination and sale contracts placed by the station and will have a right to payment of a royalty.

An agreement of this type was signed as early as 1986 between ESA and NASA in order to allow the latter to receive the SAR data at its station in Fairbanks, Alaska. Two other agreements are being discussed between ESA and Australia, and Ecuador.

2. The second level of dissemination is the use of the facilities of ESA's Earthnet programme. Established at ESRIN, Italy since 1978, the Earthnet Programme Office (EPO) is in charge of acquiring, pre-processing, archiving and disseminating satellite remote-sensing data. Until now, EPO has worked for the benefit of American and Japanese satellites prior to being used for ERS-1. EPO resorts to a network of ground stations located in Sweden, Norway, Italy and the Canary Islands. Data supplied by orbiting satellites is picked up in numerical form by any one of these stations, where it is immediately pre-processed prior to being transmitted to the users.

Earthnet has entrusted the Eurimage consortium with the responsibility for the final dissemination, *i.e.* the so-called "commercialization" of the data supplied by American Satellites. For implementing the ERS-1 programme, EPO will use the central technical facilities at ESRIN and co-ordinate the pre-processing and archiving facilities of the stations. Thus, EPO will be the central interface between programme and users and will thus constitute one of the main elements of the ground sector.

3. The third level of the ERS-1 programme consists of all the users, either individual or collective, who will have access to pre-processed or analyzed data. The Agency does not pretend to itself perform and direct the various stages of data transfer to the users; a transfer will be performed either through the national Centres in the participating States, if any, or through a specialized European group.

In this respect, ESA is now studying the possible creation of an "economic interest group" (EIG) constituted according to French law. This group would incorporate industrial and commercial companies belonging to several European countries interested in the dissemination of remote-sensing data for commercial purposes.

No final decision has yet been made and it may be envisaged that Eurimage will be given the task of disseminating and selling the ERS-1 satellite data to the users, in addition to its current tasks.

The fact that there are, at the moment in the participating States, a number of firms capable of creating some "added value" from the pre-processed data, is a factor which must be taken into account so as to be able to determine whether it is desirable that the European Space Agency should favour the arrival of new organizations on the market and whether these organizations should be granted sole rights for ERS-1 products.

Another theme of these studies is the search for a "price policy," reconciling the cost-effectiveness of the ERS-1 operation with the rivalry of the products supplied by other satellites.

Lastly, from a legal point of view, the States participating in the ERS-1 programme will have to make sure that the technical, financial and commercial conditions under which they will decide if the products of the ERS-1 satellite are to be supplied to the public are in conformity with the principles laid down in the United Nations' Resolution 41/65 of 4 December 1986. This question deserves particular attention. As a matter of fact, all of the participating States agreed to these principles and not abiding by them would be likely to require an assertion of the international responsibility of the Agency to remind the States as to their own responsibility.

CONCLUSION

The "commercialization" of remote sensing data must be considered as an irreversible, though recent, characteristic of space activities throughout the world. It appears to be a logical consequence of the attitude adopted by the United States as soon as it entered the remote-sensing field, as it claimed to be in favour of the "open sky policy" so as to widely disseminate the data collected by its satellites - other than the military ones, of course. Urged by competition considerations, the Soviet Union is trying to attract customers by offering images with a better resolution than the US ones. The newcomers in this field cannot do less from now on and they must propose products of a still higher quality, with even more advantageous financial and other conditions.

This situation explains why the European Space Agency, when embarking on its development of a remote-sensing satellite, had to consider its operation - including data dissemination - from a "commercial" viewpoint.

Actually, the various activities that can be classified in the category of "remote-sensing activities" are not necessarily "commercial" activities. They could be perfectly carried out partly by a State, on its own behalf, the remaining part being the responsbility of a non-governmental firm.

Thus, the United States would like from now on to entrust the research and development phase as well as the construction to Eosat, the private company which is already responsible for data processing and dissemination.

On the other hand, the French Spot-Image company, which was established to carry out activities in connection with data processing, seems to have limited its action to such activities.[7]

It is, as we have put it before, the fact that the Agency is entrusted with activities in connection with the operational phase which constitutes the most striking feature of the legal construction on which the ERS-1 programme has been built as an ESA optional programme.

This construction should be complemented, as stipulated in the Implementing Rules, by the adoption of additional rules concerning, in particular, the "data dissemination policy." Given that this task has not yet been accomplished as of the time of this article, we must limit ourselves to the following comments on this subject:

1. The "data dissemination policy" must be consistent with the principles stated in the United Nations' Resolution 41/65 of 1986. Some of these principles are purely political since they refer to objectives which should be those of the States which carry out remote-sensing activities. Others are more practical and refer to the conditions in which remote-sensing activities must take place. Others aim at compliance with the rights of the observed States. It seems that the principles classified in the last two categories should be specially taken into consideration in the definition and implementation of the policy to be applied in the disseminaton of ERS-1 data.

2. The participating States must also ensure the strict implementation of measures intended for the protection of their own interests and those of ESA in terms of intellectual property. For that purpose, they must take into account the principles already stated in Article 7 of the Implementing Rules and add all other principles they would deem necessary.

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7. The development and launch of ERS-2 is provided for in the initial Declaration of 24 March 1982. Recognizing the necessity of insuring the continuity of service for the benefit of users, the participants had come to an agreement on the mission of the 2nd flight model and on its characteristics with a view to its launch by Ariane in 1993. However, we learned in early 1988, that both France and the United Kingdom had decided not to participate in the extension of the ERS-1 programme. As it was not possible to gather sufficient funds in the absence of those two countries, no decision can be made at the present time on the extension of the programme.

3. It is also necessary to take into account the national laws governing space activities, which exist currently in the participating States and particularly those of the territory on which the activities of the entity in charge of data dissemination will be carried out.[8]

4. Lastly, the so-defined policy must be effectively implemented by the entity in charge of data dissemination. Thus, this entity must undertake to see that the policy will be implemented in accordance with the license contract or agreement under which ESA will entrust it with this task, and to transfer it to each final user. For its part, the Agency must provide itself with the necessary means to check that this commitment is complied with.

8. Apart from the United States, the only countries having a proper "national Space legislation" are the United Kingdom and Sweden.